

Wonsun You

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EDUCATION

Vanderbilt University

Expected May 2028

B.E. in Mechanical Engineering | Minor in Digital Fabrication (GPA: 3.55/4.00)

Nashville, TN

- **Achievements:** Dean's List: 2025, 2026
- **Coursework:** Thermodynamics, Fluid Mechanics, Mechanics of Materials, System Dynamics, Machine Analysis & Design

TECHNICAL SKILLS

- **Mechanical Design & Manufacturing:** SolidWorks (Certified SOLIDWORKS Associate; part modeling, assemblies, engineering drawings, GD&T / ASME Y14.5), CNC machining, 3D printing, welding, cutting/grinding, assembly
- **Engineering Analysis & Simulation:** SolidWorks Flow Simulation (CFD), MATLAB, basic structural and thermal/fluid analysis
- **Experimental & Optical Systems:** UV laser alignment, Raman spectroscopy, optical beam-path optimization, signal acquisition, experimental troubleshooting, ADXL335 accelerometers, NI USB-6341 DAQ, LabVIEW
- **Programming & Data Analysis:** Python (Cantera, Matplotlib, PySide6), MATLAB, Java, Microsoft Office, Git/GitHub

WORK EXPERIENCE & ACTIVITIES

Laser Diagnostics of Combustion Laboratory | *Research Assistant*

Dec 2025 – Present

- Optimized a 266 nm UV Raman spectroscopy system for high-temperature H₂ combustion, improving spectrometer coupling and Raman signal quality by 50%.
- Conducted Raman diagnostics and experimental testing on hydrogen flames up to ~2,700 K, including optical alignment, data acquisition, and system troubleshooting.
- Guided 1 undergraduate researcher through H₂ combustion experiments, Raman spectroscopy, optical setup, and data collection.

Vanderbilt University School of Engineering | *Research Intern*

May 2026 – Aug 2026

- Developed a Python/PySide6 H₂ Raman temperature-fitting tool for NASA NTREES propulsion research, estimating gas temperature from experimental spectra.
- Enhanced a physics-based spectral model with automated wavelength-shift and Gaussian-broadening optimization, achieving closer agreement between simulated and experimental spectra than the prior approach.
- Automated 3-parameter fitting and gas-composition calculations across 8 equivalence ratios ($\Phi = 1.4\text{--}4.0$) and presented results at a research symposium.

VU Motorsports - Formula SAE | *Chassis Team*

Jan 2025 – May 2026

- Fabricated and fitted chassis components using cutting, sanding, drilling, and milling to meet frame geometry and assembly requirements.
- Collaborated across chassis and vehicle subsystems during design reviews, fabrication, and integration.
- Adjusted and reworked components to improve fit, manufacturability, and final vehicle assembly.

Vanderbilt University Civil Engineering | *Welding Teaching Assistant*

August 2026 – October 2026

- Taught MIG welding fundamentals including machine setup, torch positioning, travel speed, and shop safety.
- Supervised welding labs and coached students on joint prep, weld execution, and common defect correction.

PROJECTS

High Power Rocketry - Level 1 Certification Project | *Nashville, TN*

July 2026 – Present

- Designed and simulated a Level 1 high power rocket in OpenRocket, evaluating stability, center of pressure/gravity, motor performance, and predicted flight trajectory.
- Modified a rocket kit for improved fit and assembly, including CNC-cutting replacement fins and preparing structural and recovery components.
- Preparing the vehicle for certification flight testing, with final validation of stability, recovery performance, and launch safety still in progress.

Gear-Driven Vehicle | *Nashville, TN*

Aug 2025 – Dec 2025

- Modeled the complete vehicle assembly in SolidWorks, including chassis, drivetrain, motor, axles, wheels, and gear interfaces, and produced engineering drawings following ASME Y14.5 conventions
- Designed and fabricated a 162.8 g gear-driven vehicle using a 45-tooth/11-tooth gear pair and iterative fit/alignment adjustments.
- Integrated the mechanical and electrical assembly, including soldered wiring connections, and achieved 10 ft of travel in 1.8 s during final performance testing